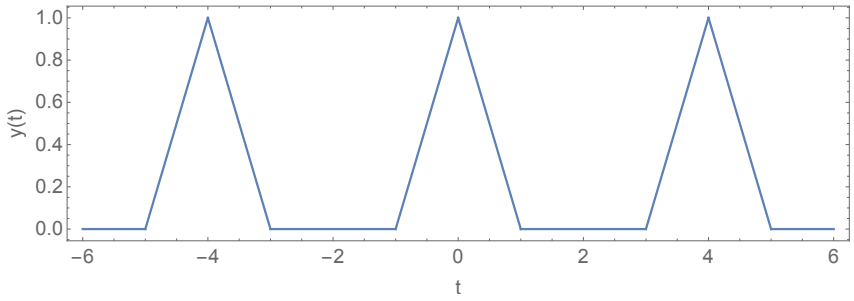
Lecturers: Thomas Quella and Nick Beaton
School of Mathematics and Statistics

Second Semester, 2026
University of Melbourne

MAST20030: Differential Equations
Tutorial: Week 8

Question 1

Consider the following function where 3 periods are shown:



(a) Determine L, (the functions period should be $2L$).

(b) Write down its algebraic form from $-L$ to L, using both Heaviside and piecewise functions.

(c) Find its Fourier series representation.

(d) How do the coefficients scale for large n?

(e) Determine whether the Fourier series converges uniformly and, if it does, prove this using the 'Weierstrass test for uniform convergence'.

Question 2

Recall from lectures that the Fourier series of x, defined for $-1 \leqslant x \leqslant 1$, converges pointwise to x if $-1 < x < 1$:

$$\frac{2}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n}\sin(\pi n x), \qquad -1 < x < 1. \tag{1}$$

We note that the Fourier coefficients decay as $\dfrac{1}{n}$, as $n \to \infty$.

(a) Using $x^2 = \displaystyle\int_0^x 2t\,\mathrm{d}t$ and (term-by-term) integration of (1), compute the Fourier series of x^2 on $-1 \leqslant x \leqslant 1$ and note that the Fourier coefficients decay as $\dfrac{1}{n^2}$, as $n \to \infty$. How do we determine the constant term?

(b) Apply a similar procedure to determine the Fourier series of x^3 on $-1 \leqslant x \leqslant 1$. Note that integration doesn't quite give a Fourier series! Find α so that the Fourier coefficients of $x^3 - \alpha x$ decay as $\dfrac{1}{n^3}$, as $n \to \infty$.

(c) Find β so that the Fourier coefficients of $x^4 - \beta x^2$ decay as $\dfrac{1}{n^4}$, as $n \to \infty$. Use the Fourier series of $x^4 - \beta x^2$ to determine the value of

$$\sum_{n=1}^{\infty}\frac{1}{n^4}.$$